Volume I · Chapter 1 — Python for Data Science (NumPy, Pandas, Matplotlib) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-I-Chapter-1-context.md` for the AI interaction log.

Prerequisites: basic Python (entry skill for the program — variables, control flow, functions, lists/dicts).

Learning outcomes — the student can:

- Manipulate numerical data efficiently with **NumPy** (vectorization, broadcasting, indexing, aggregation).
- Load, clean, transform, and join tabular **clinical data** with **Pandas** (missing values, types, dates, categoricals, merges, groupby).
- Perform basic **feature engineering** to build a modeling-ready table from raw EHR-style data.
- Produce clear exploratory **visualizations** with Matplotlib (distributions, cohort comparisons, missingness).
- Work in **reproducible, well-structured notebooks** with sound data hygiene.

Topics (theory) — 12 h:

#	Topic	h
1.1	Python for data work: environment (Anaconda/conda), Jupyter, reproducibility	1
1.2	NumPy: `ndarray`, vectorization, broadcasting,	3

#	Topic	h
	indexing/slicing, aggregation	
1.3	Pandas I: `Series`/`DataFrame`, I/O (CSV/Parquet), indexing & selection	2
1.4	Pandas II: cleaning — missing data, dtypes, dates, categoricals, merges/joins, `groupby`	3
1.5	Feature engineering for clinical/tabular data	1
1.6	Matplotlib (and pandas plotting): distributions, correlations, time series	1
1.7	Reproducible, well-structured notebooks & data hygiene	1

Datasets/tools: synthetic EHR CSVs (or a Synthea export); NumPy, Pandas, Matplotlib, Jupyter; conda/Anaconda environment.

Assessment: notebook deliverable — a cleaned dataset + short EDA report (rubric, **60%**); quiz on NumPy/Pandas idioms (**20%**); feature-engineering task producing a modeling table (**20%**).

Key decisions taught here:

- NumPy vs. Pandas for a given operation (raw arrays vs. labeled tables).

- **Wide vs. long** data format for clinical time series.
- Handling missing clinical data — **impute vs. drop** — and its **clinical implications** (bias, leakage).
- Reproducibility choices (environment pinning, deterministic notebooks).

References: plan §13 → *Programming & data science*.

Hours: Theory **12** + Lab **18** = **30**. *(Feeds plan Lab 1 — EHR Data Wrangling.)*